Lecture 12
- Particle Filter (PF)

Recall Importance Sampling

given samples $x^i \sim q(x)$
and weights $w^i = \frac{p(x^i)}{q(x^i)}$
we can approximate unknown dist. $p(x)$ by particle set $\{x^i, w^i\}_{i=1}^N$
such that

$$E_p[f(x)] \approx \sum_{i=1}^N f(x^i) w^i$$

graphically



Q: How is this related to state est.?
A: Gives a simple method for approx. Bayes' Rule:

$$p(x|y) = \frac{p(y|x)\, q(x)}{p(y)}$$

proposal dist
target dist

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claim: We can get w^i without knowing $q(x)$ or $p(x|y)$!

recall weighting function

$$w(x) = \frac{p(x|y)}{q(x)} = \frac{p(y|x)}{p(y)}$$

↑ Bayes' Rule

$w(x) \propto p(y|x)$
↑ proportional to

so given $Y = y$, $X = x^i$, $X \sim q(x)$ prior

unnormalized: $\hat{w}^i = p(Y = y \mid X = x^i)$

normalized: $w^i = \frac{\hat{w}^i}{\sum_{j=1}^N \hat{w}^j}$

and posterior $p(x|y)$ is represented by $\{x^i, w^i\}_{i=1}^N$

Particle Filter Algorithm

(Sequential Importance Resampling, SIR Filter)

Init: particle set, uniform weights

$$\{x_{0|0}^i, w_0^i = \tfrac{1}{N}\}_{i=1}^N, \quad x_{0|0}^i \sim p(x_0) \text{ i.i.d.}$$

predict:

$$x_{t|t-1}^i = f(x_{t-1|t-1}^i, u_{t-1}) + w_{t-1}^i$$

where $w_{t-1}^i \sim p_w(w_{t-1})$ process noise dist.

predict each particle with different random noise draw

(no change to weights)

update:

$$\hat{w}_t^i = p(Y_t = y_t \mid X_t = x_{t|t-1}^i)$$

find measurement likelihood for each particle $x_{t|t-1}^i$

$$w_t^i = \frac{\hat{w}^i}{\sum_{j=1}^N \hat{w}^j}$$

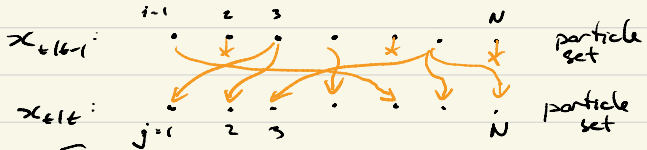
normalize

importance resample:

$$\{x_{t|t}^j, \underbrace{w_t^j = \tfrac{1}{N}}_{\text{uniform weights}}\}_{j=1}^N \Leftarrow \{x_{t|t-1}^i, w_t^i\}_{i=1}^N$$

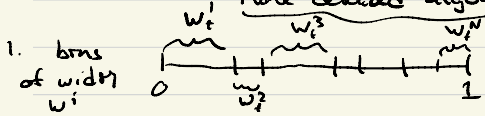
Importance Resampling (IR)

main idea: choose a new set of particles, $x^j_{t|t}$ by selecting $x^i_{t|t-1}$ with prob. w^i_t.



for $j=1$ to N
  $x^j_{t|t}$ draw from $\{x^i_{t|t-1}\}_{i=1}^N$
    where $x^j_{t|t} = x^i_{t|t-1}$ with prob w^i_t
end
set $w^j_t = \frac{1}{N}$ $\forall j = 1, \dots, N$

More detailed algorithm for IR



2. draw random scalar s_j uniformly on $[0,1]$
3. if s_j lands in bin i, assign $x^j_{t|t} = x^i_{t|t-1}$
4. repeat for $j = 1, \dots, N$.

Practical Tips

(i) Usually noise is Gaussian
$W^i_t \sim N(0, Q_t)$
$V^i_t \sim N(0, R_t)$
but doesn't have to be!

(ii) Notice we require
(a) "generative" model for
$W^i_t \sim p_W(w_t)$
to draw samples
(b) "discriminative" or analytic model for likelihood:

$$\hat{w}^i_t = p(Y_t = y_t \mid X_t = x^i_{t|t-1})$$

E.g. consider Gaussian sensor noise with nonlinear sensor model

$$Y_t = g(X_t) + V_t, \quad V_t \sim GWN(0, R_t)$$

notice: $V_t = Y_t - g(X_t)$

and $V_t \sim \eta\, e^{-\frac{1}{2} v_t^T R_t^{-1} v_t}$

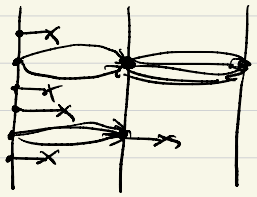
so $(Y_t - g(X_t)) \sim \eta\, e^{-\frac{1}{2}(y_t - g(x_t))^T R_t^{-1} \times (y_t - g(x_t))}$

$= p(y_t \mid x_t)$

to compute $p(Y_t = y_t \mid X_t = x^i_{t|t-1})$
we plug in:
$\eta\, e^{-\frac{1}{2}(y_t - g(x^i_{t|t-1}))^T R_t^{-1}(\dots)}$

(iii) Failure Modes

(a) Particle Impoverishment



cause: resample too often

solution: resample less often

$\{x^i_{t|t-1}, w^i_t\}_{i=1}^N$
↳ no resample
$\{x^i_{t|t}, w^i_t\}_{i=1}^N$
$\|$
$x^i_{t|t-1}$ (keeps weighted particles)

at update:

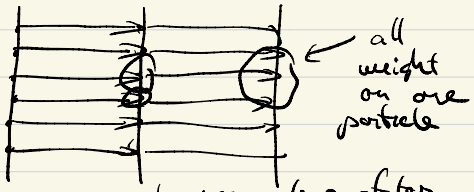
$$\hat{w}^i_{t+1} = p(Y_{t+1} = y_{t+1} \mid X_{t+1} = x^i_{t+1|t}) \cdot w^i_t$$

$$w^i_{t+1} = \frac{\hat{w}^i_{t+1}}{\sum_{i=1}^N \hat{w}^i_{t+1}}$$

proceed.

(b) Particle Degeneracy

opposite problem



cause: not resampling often enough.

tests for these failure modes:

(a) if $\frac{1}{N}\sum_{i=1}^{N} \| x_{t|t}^{i} - \mu_{t|t} \|^2 \leq$ thresh

unweighted average

then don't resample

(b) if $\frac{1}{N}\sum_{i=1}^{N} (w_t^i - \frac{1}{N})^2 \geq$ thresh

then do resample